SYED SHAIL SUHEB

Dubai, UAE | +971 54 462 1169 | syedshailsuheb@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI/ML-focused Software Engineer with hands-on experience building machine learning, recommendation, NLP, and retrieval systems. Experienced in Python, Scikit-learn, Random Forest, vector embeddings, FAISS, LLMs, RAG, Flask, and cloud technologies. Built end-to-end AI systems involving natural-language query understanding, semantic retrieval, structured filtering, and ML ranking. Strong software engineering foundation across backend development, databases, APIs, and cloud deployment. Incoming MSc Artificial Intelligence and Machine Learning student at the University of Birmingham.

TECHNICAL SKILLS

AI / Machine Learning: Machine Learning, Scikit-learn, Random Forest, K-Means, NLP, RAG, LLMs, Vector Embeddings, Semantic Search, FAISS, Pandas, NumPy, TensorFlow

AI Engineering: Recommendation Systems, ML Ranking, Retrieval-Augmented Generation, Prompt Engineering, Vector Databases, LLM Integration, Model Evaluation

Programming: Python, C++, JavaScript, TypeScript, Java, SQL

Backend / Data: Flask, FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, Firebase, Redis, ChromaDB

Cloud / Tools: AWS (EC2, S3, Lambda, RDS), Docker, Git, GitHub Actions, LangChain, Tree-sitter, Jupyter, Linux/Unix

WORK EXPERIENCE

Software Engineering Intern – Wipro Limited Oct 2025 – Present

Bangalore, India

- Developed a **TypeScript-based VS Code extension** for automated cryptographic code analysis across C++, Python, and Java repositories, processing **70+ lines/sec** across large codebases.
- Built an AST-based analysis pipeline using **Tree-sitter** to parse source code and detect cryptographic algorithms and security-relevant patterns including **RSA, AES, and SHA**.
- Designed automated cryptographic inventory generation using **CycloneDX 1.6 CBOM**, enabling organizations to identify cryptographic dependencies and support migration toward post-quantum algorithms.
- Implemented asynchronous repository scanning and caching mechanisms, reducing analysis time by **60%** and improving scalability across large multi-language repositories.

PROJECTS

CarSense – AI Vehicle Recommendation Platform *Python, Flask, Scikit-learn, FAISS, LLMs, React, PostgreSQL, Docker, AWS*

- Built an end-to-end **hybrid AI recommendation system** combining LLM-based natural-language query understanding, vector embeddings, FAISS semantic retrieval, structured filtering, and machine-learning-based ranking.
- Developed an **LLM query understanding pipeline** that extracts user intent, budget, seating capacity, vehicle type, fuel preference, reference vehicles, and comparative requirements from natural-language queries.
- Implemented **FAISS vector search** to retrieve semantically relevant vehicles, followed by hard constraints and a trained **Random Forest ranking model** to score and prioritize candidates.
- Built a ranking pipeline that combines semantic relevance with structured vehicle attributes, enabling personalized recommendations for complex queries such as finding cheaper alternatives with similar capabilities.
- Developed Flask APIs and PostgreSQL data pipelines and integrated Docker and AWS infrastructure, supporting a scalable architecture designed for **10K+ daily requests**.

AI Study Assistant with RAG *Python, LangChain, ChromaDB, OpenAI API, Streamlit, Docker*

- Developed a **Retrieval-Augmented Generation (RAG)** application for document-based question answering using LangChain, vector embeddings, ChromaDB, and semantic retrieval.
- Engineered a document ingestion pipeline supporting **PDF, DOCX, and TXT** files with document chunking, metadata extraction, embedding generation, and vector storage across **1K+ pages**.
- Improved answer relevance through prompt engineering and few-shot learning while providing source citations and chat history through a Streamlit interface.

JSS Tax Management System *React, Node.js, Express.js, PostgreSQL*

- Built an enterprise tax management platform handling **500+ daily submissions**, reducing manual effort by **70%** and processing time by **80%**.
- Optimized Express.js services and PostgreSQL queries using connection pooling, indexing, and query tuning, achieving sub-**150ms** response times and reducing database query time by **60%**.
- Implemented automated validation and workflow processing to improve data accuracy by **40%**, alongside secure authentication, session management, and audit logging.

EDUCATION

University of Birmingham Sep 2026 – Aug 2027

MSc Artificial Intelligence and Machine Learning *Birmingham, UK*

JSS Science and Technology University 2022 – 2026

B.E. Computer Science and Business Systems – GPA: 8.6/10.0 *Mysore, India*

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning, Artificial Intelligence, Cloud Computing, Cybersecurity